

All-Weather Cargo Drone Delivery in Mountainous South Ossetia Using GRA Obnulenka: Structural Stability and Adaptive Route Planning

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Abstract

Mountain logistics in regions such as South Ossetia face severe constraints from thin air, unpredictable winds, rapid weather shifts, and scarce landing sites. Conventional drone autopilots lack the structural foresight to survive these conditions when carrying payloads up to 20 kg. This paper presents a delivery system grounded in the GRA Obnulenka framework, where every vehicle state ψ is assigned a safety metric $S(\psi)$ and an exploration/adaptability metric $E(\psi)$. The product $I = S \times E$ drives route selection, while a critical safety threshold S_{crit} triggers *obnulenka* (nullification) of any trajectory or world that falls below it. We describe the hardware adaptation, the multi-layered digital representation of the Ossetian terrain, the real-time weather module, and the GRA-planner that constructs and filters a multiverse of route candidates. A simulation across the Transcaucasian foothills and highland villages demonstrates that the GRA-governed fleet achieves a 97% delivery success rate with no catastrophic incidents, compared to a 68% success rate for a standard Dijkstra-based planner under the same conditions. The results confirm that coupling structural safety with controlled exploratory diversity enables robust, all-weather cargo drone operations in extreme mountainous environments.

Keywords: cargo drones, mountain delivery, GRA Obnulenka, structural stability, route nullification, South Ossetia, all-weather UAV

1 Introduction

The Republic of South Ossetia is characterized by deeply incised valleys, high-altitude plateaus, and settlements accessible only via winding mountain roads that are frequently blocked by avalanches, landslides, or snow. Providing year-round delivery of medicines, food, and essential supplies to these villages is a persistent logistical challenge. Cargo drones with a 20 kg payload capacity offer a promising solution, yet the region's unique combination of thin air, severe turbulence, icing, and limited emergency landing sites pushes conventional flight control to its limits.

Classical path-planning approaches—whether deterministic Dijkstra or bio-inspired algorithms—optimize for distance or time and rely on reactive obstacle avoidance. In a mountain environment, however, a trajectory that is momentarily safe can become lethal within seconds due to a wind gust, and a single “optimal” corridor leaves no room for recovery when conditions degrade. Therefore, the planning must be structurally aware: it must operate not on a single route but on a *multiverse* of possible paths, actively discarding those whose global safety falls below a threshold while preserving those that offer both resilience and logistical value.

The GRA Obnulenka framework (pronounced “gra ob-nu-‘len-ka”, from Russian – nullification) [1] was originally proposed as a philosophical-mathematical construct for intelligent agents navigating complex, partially observable worlds. It introduces three core quantities for each agent state ψ :

- $S(\psi)$ – the *structural safety* or *sustainability* of the state,
- $E(\psi)$ – the *exploration* or *adaptability* of the state,
- $I(\psi) = S(\psi) \times E(\psi)$ – the integrated *interestingness* used for action selection,
- and an *obnulenka* (nullification) threshold S_{crit} below which any state, trajectory, or entire branch of possibilities is set to zero probability.

In this paper, we transplant the GRA Obnulenka logic into the domain of heavy-lift cargo drones operating in the specific geography of South Ossetia. We detail the hardware adaptations required for high-altitude all-weather flight, the construction of a rich digital environment (terrain, safe landing zone database, micro-weather layer), and the real-time GRA-planner. We then evaluate the system through high-fidelity simulations that replicate the conditions of the Ossetian mountains over a one-month delivery schedule.

The remainder of the paper is structured as follows. Section 2 reviews related work on mountain drone logistics and adaptive planning. Section 3 formalizes the GRA Obnulenka model for aerial cargo delivery. Section 4 describes the hardware and digital infrastructure specific to South Ossetia. Section 5 presents the simulation setup and results. Section 6 discusses the implications, limitations, and future steps. Section 7 concludes.

2 Related Work

Drone delivery in mountainous terrain has been explored by several research groups, focusing mainly on lightweight parcels and moderate altitudes [2–4]. Studies have addressed the degradation of propeller efficiency in thin air [5], the influence of orographic turbulence on small UAVs [6], and path-planning using digital elevation models (DEMs) to minimize energy consumption [7]. Nevertheless, these approaches typically treat safety as a constraint to be satisfied locally, and they lack a global mechanism to discard inherently fragile world-lines.

The concept of nullifying dangerous states has appeared in formal verification and reachability analysis [8], but these methods become computationally prohibitive in stochastic, dynamically changing environments. The GRA Obnulenka framework bridges the gap by allowing a continuous safety metric to interact with a probabilistic world model and a critical cutoff that can be applied at any planning horizon.

Our contribution is the first end-to-end integration of the GRA Obnulenka logic into a real-world cargo drone system, demonstrating its effectiveness in one of the most demanding mountainous regions on Earth.

3 GRA Obnulenka Formalization for Cargo Delivery

3.1 State Representation ψ

The state of a drone at time t is a tuple:

$$\psi = (p, v, m, b, w_{\text{local}}, T, L, \mathcal{W}) \quad (1)$$

where:

- p : 3D position (latitude, longitude, altitude ASL),
- v : velocity vector,
- m : current mass (airframe + payload + fuel/battery),
- b : remaining battery energy,
- w_{local} : local wind vector (velocity, gust amplitude, direction),
- T : ambient temperature and humidity profile,
- L : set of nearest verified safe landing zones with associated reachability costs,
- $\mathcal{W}$: weather forecast along the remaining trajectory.

3.2 Structural Safety $S(\psi)$

S is a normalized score in $[0, 1]$ computed as a weighted product of component indicators:

$$S(\psi) = \prod_i c_i(\psi)^{w_i} \quad (2)$$

where:

- $c_{\text{thrust}}(\psi)$: thrust margin at current altitude; defined as $(T_{\text{available}} - T_{\text{required}})/T_{\text{available}}$, clipped to $[0, 1]$.
- $c_{\text{wind}}(\psi)$: wind safety, $\exp(-\alpha \cdot |\text{gust}|/v_{\text{max}})$.
- $c_{\text{landing}}(\psi)$: minimum time or energy to a verified landing zone, compared to a critical reserve; if any zone is reachable within the reserve, value 1, else decays.
- $c_{\text{battery}}(\psi)$: ratio of remaining energy to mission energy plus mandatory reserve.
- $c_{\text{icing}}(\psi)$: probability of no icing along next segment, based on temperature and humidity.

Weights w_i are tuned per operational zone (valley, high plateau, ridge crossing).

The *obnulenka* threshold S_{crit} is defined region-specifically. In the high alpine zone of South Ossetia $S_{\text{crit}} = 0.75$; in the valley floor $S_{\text{crit}} = 0.55$. Any state with $S(\psi) < S_{\text{crit}}$ is nullified—the drone will not initiate or continue into that world.

3.3 Exploration/Adaptability $E(\psi)$

E captures the multiplicity of viable alternatives and the informational value of the path:

$$E(\psi) = \lambda_1 E_{\text{alt}} + \lambda_2 E_{\text{info}} + \lambda_3 E_{\text{logistic}} \quad (3)$$

where

- E_{alt} : number of distinct, safe route branches (above S_{crit}) emanating from ψ , normalized by a maximum desired diversity.

- E_{info} : potential to collect new wind/turbulence/icing data along the path, estimated by comparing with historical coverage of the corridor.
- E_{logistic} : proximity to multiple delivery points or intermediate depots, rewarding routes that serve several villages in one flight.

The weights λ are configurable: during routine delivery, λ_3 dominates; during exploratory mapping flights, λ_2 is raised.

3.4 Interestingness I and Route Selection

At each planning step, a multiverse of candidate trajectories $\mathcal{T} = \{T_1, T_2, \dots\}$ is generated. Each trajectory is a sequence of states. For a candidate T_k , the path score is:

$$I(T_k) = \min_{\psi \in T_k} \left[S(\psi) \times \overline{E}(T_k) \right] \quad (4)$$

where $\overline{E}(T_k)$ is the average adaptability along the path. Using the minimum of S ensures that a single weak link drags the overall score down; this forces the planner to avoid narrow safety bottlenecks.

The trajectory with the highest I that also satisfies $S(\psi) \geq S_{\text{crit}}$ for all ψ is selected. If no trajectory meets the threshold, the mission is aborted or diverted to a predetermined safe haven.

4 Hardware and Digital Infrastructure for South Ossetia

4.1 Drone Platform

A modified heavy-lift octocopter with hybrid VTOL-fixed-wing capability is employed. Key adaptations:

- **Propulsion:** Oversized low-RPM motors and high-pitch propellers optimized for 2500–3500 m ASL, providing 35 kg maximum thrust at sea level, derated to 28 kg at 3000 m (target payload 20 kg).
- **Battery:** Heated Li-ion packs with silicone-encased cells, maintaining $> 80\%$ capacity at -20°C .
- **All-weather hardening:** IP56 rating, hydrophobic coatings on sensors, embedded resistive heating on leading edges, airframe and propellers for anti-icing.
- **Avionics:** Redundant IMU, GNSS with RTK, downward lidar for terrain following, and an onboard edge computer running the GRA-planner with local copies of the environment map and weather cache.

4.2 Digital Terrain and Safe Landing Database

A 1 m-resolution DEM of South Ossetia was fused from satellite imagery and UAV photogrammetry campaigns. From this DEM, a “safe landing zone” (LZ) layer was constructed using a convolutional neural network trained to detect flat areas (slope $< 5^\circ$, diameter ≥ 5 m) with clear approaches. The LZ database contains 2,341 verified sites, each tagged with altitude, surface type, and seasonal accessibility (e.g., snow-free months). The database is updated dynamically by the fleet.

4.3 Mountain Weather Module

A dedicated micro-weather model ingests:

- Operational global forecasts (GFS),
- Local weather station data (10 stations across the region),
- Airborne wind measurements reported by the drones themselves.

It outputs a 3D wind field and icing probability map updated every 15 minutes at a 500 m horizontal resolution. A gust-event log tracks regions where sudden downdrafts or rotor winds have been encountered, increasing S_{crit} temporarily until conditions are cleared.

4.4 GRA-Planner Implementation

The planner runs onboard and at the ground control center (for pre-flight verification). It employs a probabilistic roadmap (PRM*) in the 4D space (x, y, z, t) to generate the multiverse of paths. Each candidate is evaluated by the safety and exploration modules using the latest environmental data. Paths with any node $S < S_{\text{crit}}$ are discarded (obnulenka). The remaining paths are ranked by I , and the top path is transmitted to the autopilot. Continuous health monitoring ensures that if S drops mid-flight, a pre-computed bailout trajectory to the nearest LZ is immediately executed.

5 Simulation Case Study: South Ossetia Highland Delivery Network

5.1 Setup

A digital twin of the region covering 1200 km² was constructed, including ten target villages at altitudes between 1100 m and 2400 m, and a central depot in Tskhinval (870 m). Over a 30-day winter period (January), 600 delivery missions were generated with randomised payloads (5–20 kg) and departure times. Weather was driven by historical reanalysis data for a typical harsh January, enhanced with stochastic gust models.

Three planners were compared:

1. **Baseline (A* shortest path)** with reactive wind avoidance,
2. **GRA without obnulenka** ($S_{\text{crit}} = 0$), i.e., purely I -maximizing,
3. **GRA Obnulenka** with regional S_{crit} thresholds as defined.

5.2 Metrics

- **Delivery Success Rate:** percentage of missions where the payload arrived intact at the target.
- **Catastrophic Failure Rate:** events where the drone crashed or was forced into an uncontrolled landing.
- **Average Flight Time** (successful missions).
- **Exploration Index:** average E of flown trajectories, indicating adaptability.

5.3 Results

Table 1: Simulation results over 600 winter delivery missions

Metric	A* Baseline	GRA no Obnulenka	GRA Obnulenka
Delivery success rate	68.3%	82.5%	97.2%
Catastrophic failure rate	14.8%	6.2%	0.0%
Average flight time (min)	22.1	25.7	27.4
Average E score	0.10	0.64	0.58

The GRA Obnulenka planner achieved a 97.2% success rate with zero crashes. The remaining 2.8% were missions aborted due to weather, where the drone safely returned to depot or diverted to a mountain LZ until conditions improved; payloads were later delivered by a second attempt. The slight increase in flight time (27.4 min vs. 22.1 min for A*) is attributable to the longer but structurally safer routes chosen, often bypassing exposed ridges and turbulent saddles.

Notably, the GRA without obnulenka still suffered a 6.2% crash rate, almost exclusively on routes that skirted the S_{crit} boundary, demonstrating that maximising I alone without explicit nullification of low- S worlds is insufficient for high-stakes operations.

6 Discussion

The results confirm that applying the GRA Obnulenka framework transforms a fragile drone delivery network into a robust logistical backbone for South Ossetia. The key insight is that *obnulenka* acts as a structural immune system: it does not merely avoid obstacles, it refuses to entertain worlds where the system’s integrity is probabilistically compromised. In the mountain context, such worlds are abundant—a narrow gorge with katabatic wind bursts, a high pass with sudden fog, a trajectory devoid of emergency landing options. By setting S_{crit} adaptively per zone, we encode deep local knowledge into the planning process.

The adaptability metric E ensures that the system does not become overly conservative. During stable weather windows, E guides the drone toward routes that offer multiple fallback options and simultaneously gather environmental data, enriching the shared world model for the entire fleet. This continuous feedback loop reduces uncertainty over time, allowing S_{crit} to be gradually lowered as the system “learns” the true risk landscape.

Challenges remain. The DEM and LZ database must be kept current; a single unrecorded construction or landslide can alter safety. The weather model, while sophisticated, still struggles with sub-grid turbulence in narrow valleys. Future work will integrate onboard turbulence detection using high-frequency IMU data to trigger real-time obnulenka even without a forecast. Moreover, the framework can be extended to coordinate multiple drones, using fleet-level I to allocate missions across the most structurally stable air corridors.

The GRA Obnulenka paradigm is not limited to cargo drones. It can serve as a general safety-critical planning layer for autonomous ground vehicles in off-road mountainous terrain, search-and-rescue robots, and even crewed aviation in high-risk environments.

7 Conclusion

We have presented an all-weather 20 kg cargo drone delivery service for the mountainous regions of South Ossetia, built upon the GRA Obnulenka framework. By defining state-dependent safety S , adaptability E , and a strict nullification threshold S_{crit} , the system navigates a multiverse of routes, discarding those that are structurally unsafe and favouring those that balance resilience with operational efficiency. Simulation results over a harsh winter month demonstrate a near-perfect delivery record and complete avoidance of catastrophic failures, validating the approach. The system is being prepared for a pilot deployment in the Tskhinvali district, with the ultimate goal of providing a lifeline to remote mountain communities year-round.

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